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VELALAR COLLEGE OF ENGINEERING AND TECHNOLOGY

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Department of Electronics and Communication Engineering

DESIGN AND PERFORMANCE ANALYSIS OF HIGH SPEED SYNCHRONOUS BINARY COUNTER USING PASS TRANSISTOR LOGIC

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OBJECTIVE

- To design and implement the synchronous binary counter using Pass Transistor Logic (PTL) for reducing transistor count and hardware complexity.
- To analyze and compare the performance of CMOS and PTL-based counters in terms of power consumption, propagation delay, silicon area, and speed efficiency.

LITERATURE SURVEY

Authors & Year	Title	Name of the Journal	Key Features
Sungho Kim, Jeongmin Kim, In-Cheol Park Year : 2025	High-Speed CMOS Synchronous Binary Counter With Constant Counting Rate	IEEE Access	Proposed pre-scaled architecture; constant counting rate; reduced critical path delay; achieved ~3.7 GHz
Yujin Hyun, In-Cheol Park Year : 2021	Constant-Time Synchronous Binary Counter With Minimal Clock Period	IEEE Transactions on Circuits and Systems II	Eliminated carry chain delay; constant-time operation; improved clock frequency (~2 GHz)
S. Kim et al. Year : 2021	Constant Delay Systolic Binary Counter With CA-Based Prescaler	Microelectronics / Computers & Electrical Engineering	Introduced systolic architecture; reduced delay; improved scalability and high-speed performance

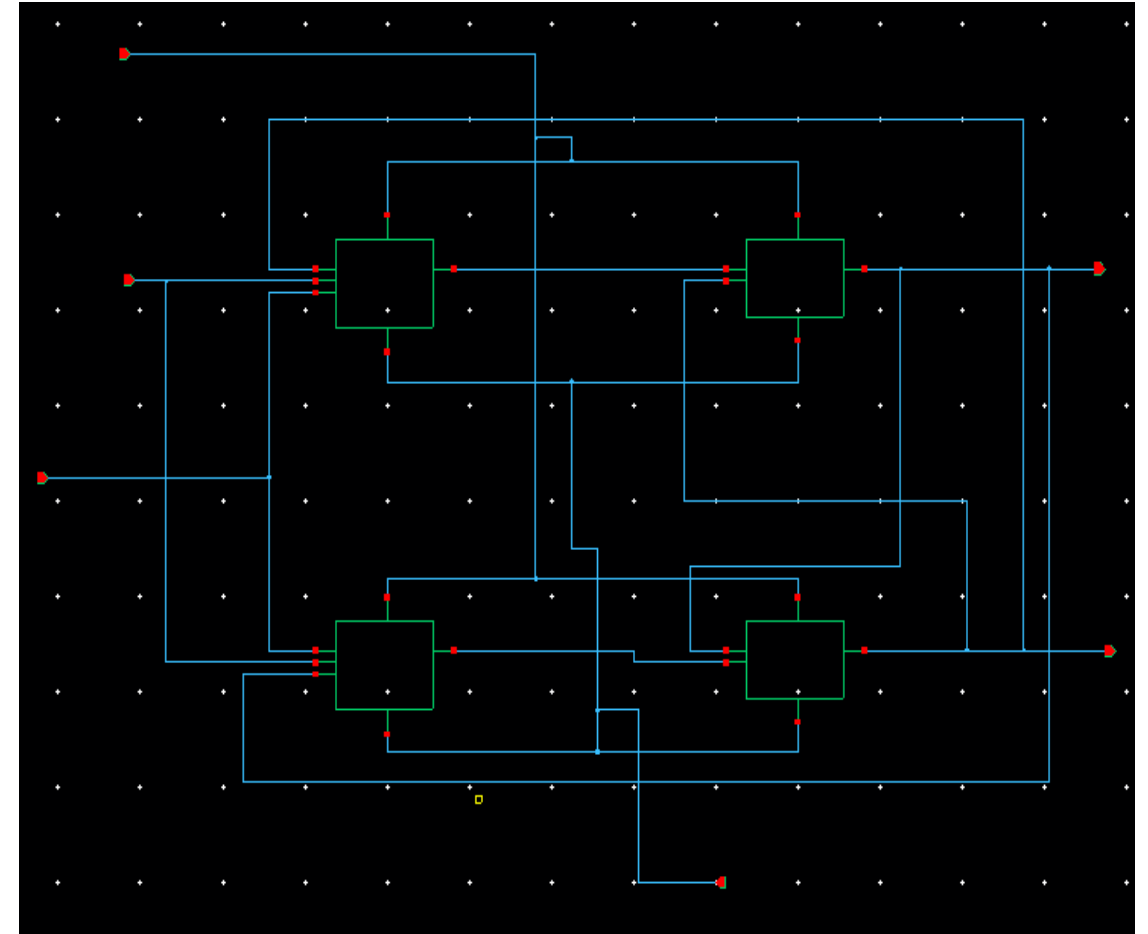
Authors & Year	Title	Name of the Journal	Key Features
R. Zimmermann, W. Fichtner Year : 1997	Low-Power Logic Styles: CMOS Versus Pass Transistor Logic	IEEE Journal of Solid-State Circuits	PTL reduces transistor count and capacitance; improves speed and power efficiency; highlights voltage degradation issue
V. G. Oklobdzija Year : 1995	Pass-Transistor Logic Versus CMOS Logic for High-Speed Low-Power Circuits	IEEE Transactions on VLSI Systems	Demonstrated PTL advantages in high-speed circuits; reduced power-delay product; suitable for efficient VLSI design

EXISTING SYSTEM

- Based on conventional CMOS logic design
- Uses flip-flops with combinational logic gates for counter operation
- Requires large number of transistors, increasing hardware complexity
- Consumes higher power due to switching activity and capacitance
- Has propagation delay due to multi-level logic and carry generation

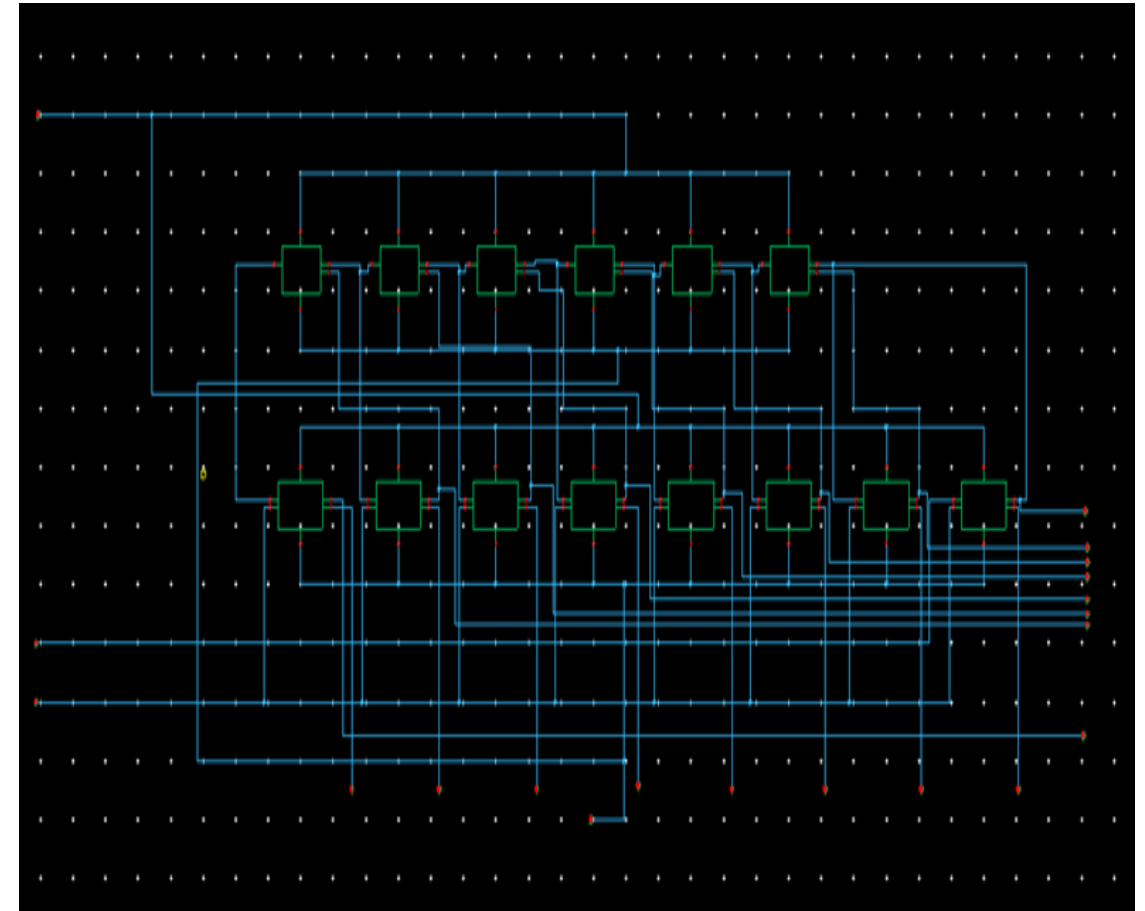
CMOS FLIP-FLOP DESIGN

- Sequential circuit used in synchronous counters
- Designed using CMOS logic
- Transistor Count: 20
- Provides stable and reliable operation
- Higher power consumption and area
- Increased hardware complexity



CMOS SYNCHRONOUS BINARY COUNTER

- Implemented using CMOS logic
- Transistor Count: 196
- All flip-flops connected to common clock
- Performs synchronized counting operation
- Higher propagation delay
- Increased power consumption and chip area

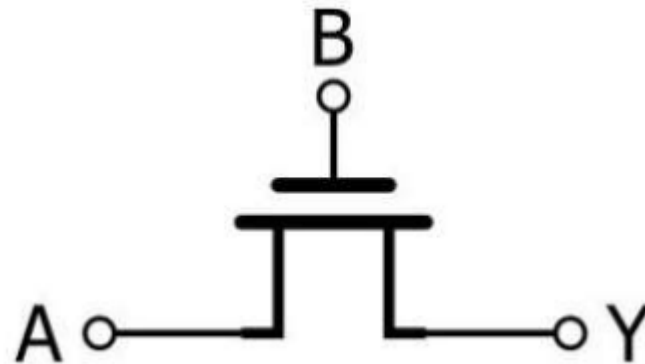


PROPOSED SYSTEM

- ◆ **Pass Transistor Logic Design** – Flip-flops and gates use PTL to reduce transistor count and improve speed.
- ◆ **Synchronous Counter Structure** – All flip-flops share a common clock for accurate counting.
- ◆ **High-Speed Operation** – Optimized PTL logic reduces propagation delay for faster performance.
- ◆ **Low Power Consumption** – Fewer transistors and lower switching capacitance reduce power usage.
- ◆ **Area Efficient Design** – Compact PTL circuits require less chip area than conventional CMOS.

PASS TRANSISTOR

- Pass Transistor Logic (PTL) reduces transistor count in digital circuits by using MOSFETs as switches to pass input signals to the output, rather than directly connecting to power rails.



- It enables faster, lower-power and compact implementations of logic gates (e.g., AND, XOR) compared to traditional CMOS, but often suffers from voltage drops ($V_{DD} - V_T$).

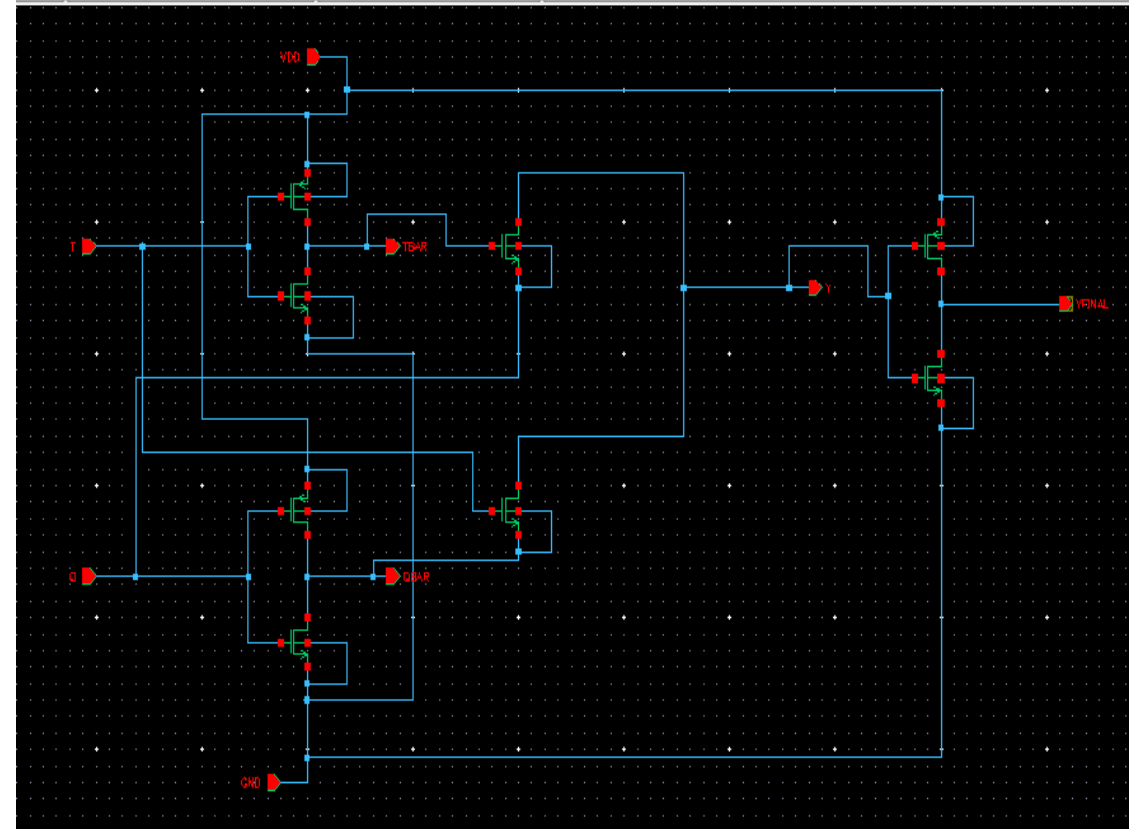
TOOLS REQUIRED

Software:

Cadence Virtuoso – Circuit design and simulation

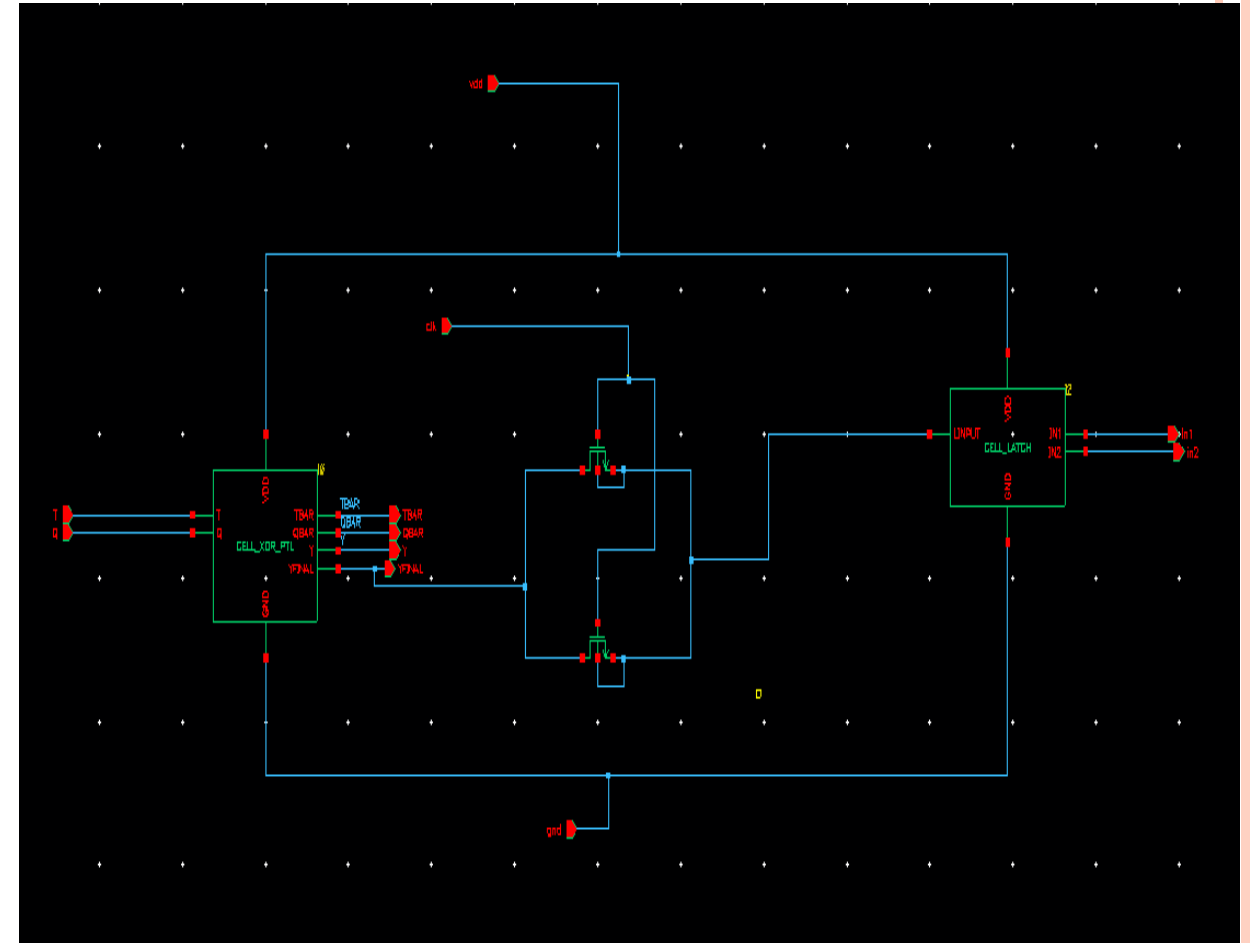
PTL XOR GATE

- Implemented using Pass Transistor Logic
- Transistor Count: 8
- Reduces circuit area and complexity
- Improves switching speed
- Lower power consumption
- Suitable for high-speed VLSI circuits



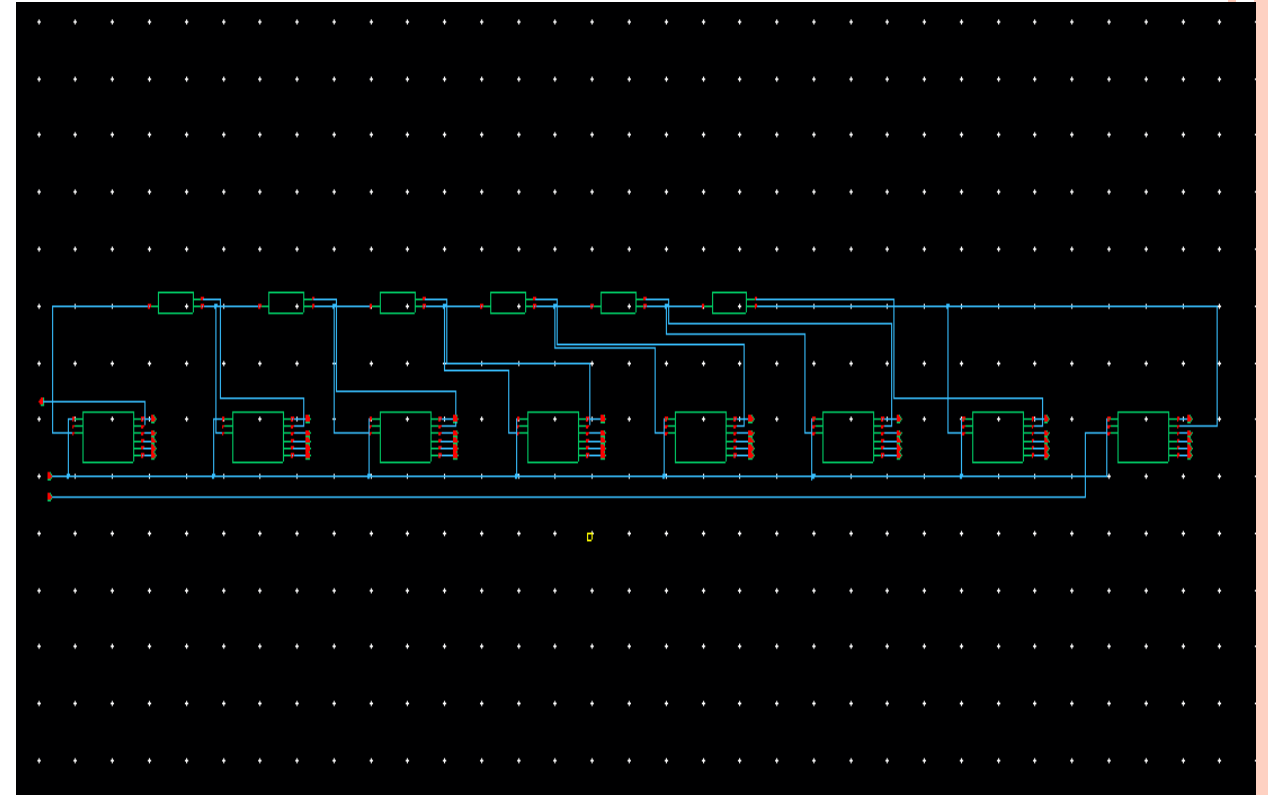
PTL FLIP-FLOP DESIGN

- Designed using pass transistor structures
- Transistor Count: 14
- Reduces Silicon area
- Lower power dissipation
- Simplified signal paths
- Improves switching speed

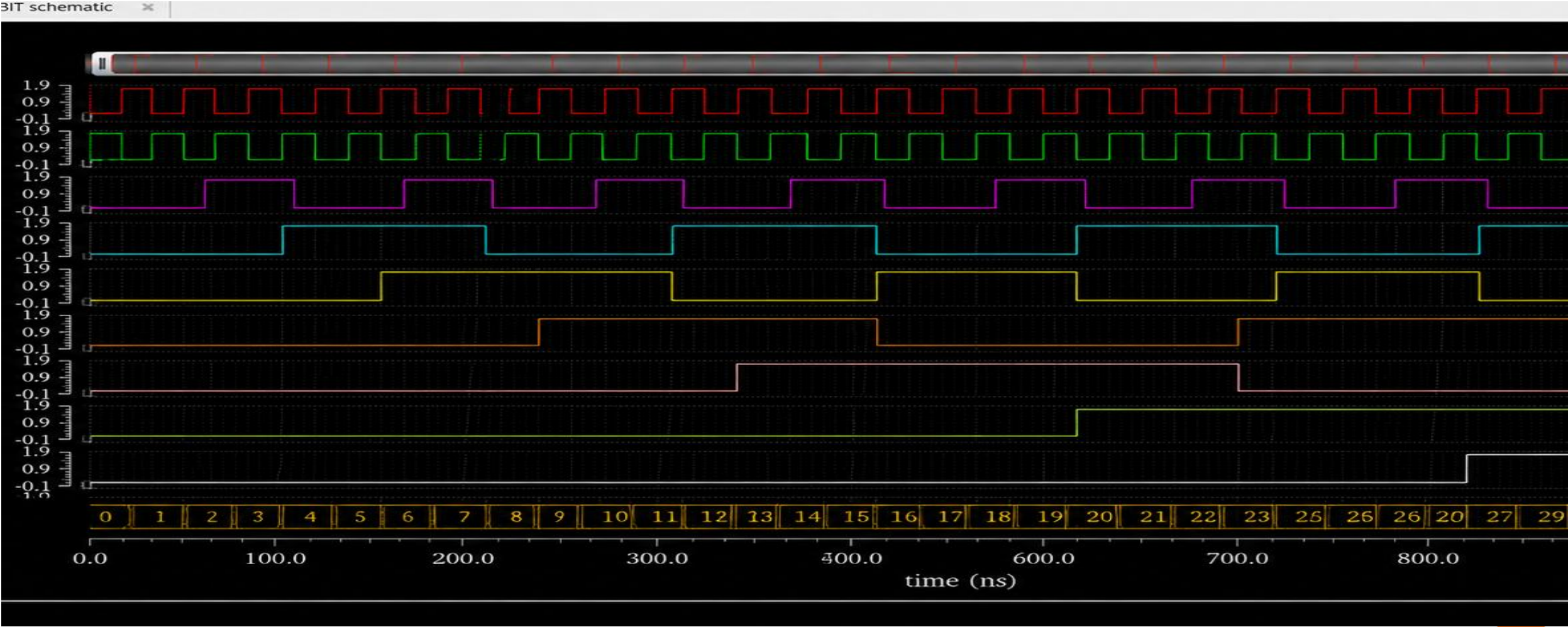


PTL SYNCHRONOUS BINARY COUNTER

- Proposed high-speed counter architecture
- Transistor Count: 124
- Uses PTL-based flip-flops
- Reduced propagation delay
- Improved counting speed
- Better hardware efficiency



PTL COUNTER WAVEFORM



TRANSISTOR COUNT COMPARISION

S. No	Parameter	CMOS	PTL
1.	AND Gate	6	2
2.	XOR Gate	12	8
3.	Flip-Flop	20	14
4.	8 Flip-Flops	160	112
5.	Binary Counter	196	124
6.	Backward Propagation Circuit	286	154

ADVANTAGES

- Reduced transistor count compared to conventional CMOS design
- Lower power consumption due to simplified circuit structure
- Reduced silicon area and improved area efficiency
- Lower hardware complexity using Pass Transistor Logic (PTL)
- Reduced propagation delay and improved operating speed
- Faster signal propagation through optimized logic paths
- Compact and efficient circuit implementation
- Suitable for low-power and high-speed VLSI systems

APPLICATIONS

- Digital counters and timing circuits
- Frequency divider circuits
- Digital clocks and stopwatch systems
- High-speed VLSI systems
- Low-power digital electronics applications
- Sequence generators and control circuits
- Embedded digital hardware systems
- Communication and signal processing circuits
- Microprocessor and memory addressing systems

CONCLUSION

This project presents the design of an asynchronous binary counter using Pass Transistor Logic (PTL). The proposed PTL design reduces transistor count, silicon area, and hardware complexity compared to CMOS logic. It also lowers power consumption due to reduced switching activity. The counter achieves improved speed performance with reduced propagation delay. Transient waveform analysis confirms the correct counting operation of the circuit. Overall, the PTL-based counter provides a compact, low-power, and high-speed VLSI design suitable for modern digital applications.

FUTURE SCOPE

- Can be extended for higher-bit counter designs
- Suitable for advanced low-power VLSI applications
- Can be implemented using modern CMOS technologies

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THANK YOU